

A Hyper-ledger Fabric Framework as a Service for Improved Quality E-voting System

Abstract—Blockchain is often referred to as a disruptive technology but it has emerged from relative obscurity to become a relatively popular conversation topic across various industries. Many sectors, like finance, medicine, manufacturing, and education, use blockchain applications to profit from the unique bundle of characteristics of this technology. Blockchain technology (BT) promises benefits in trustability, collaboration, organization, identification, credibility, and transparency. As we know, Elections are a symbol of hope and freedom, and the right to vote is an expression of belonging and of having a voice. In this paper, we'll describe about blockchain on E-voting which offers to the electoral process is a combination of accuracy, transparency, and immutability. The applications are virtually limitless, and government is certainly not the only realm within which important elections take place, where security is paramount, and where blockchain can make a difference. No matter the arena, voters deserve trust. They demand it, they deserve it. And in this paper, we proposed the most secured way of voting with the help of Blockchain using Hyper-ledger Fabric based Framework as a Service (FaaS) which can used to implement a highly maintainable, large-scale, cost effective E-voting solution within personalized private blockchain.

Index Terms—E-voting. Blockchain. Hyper-ledger. Fabric. Security. Privacy. Block. Mining. Smart. Contact.

I. INTRODUCTION

Voting has progressed in technology from traditional days when voters dropped votes marked on a shell, shard of pottery, or card into a box to the current days where voting is controlled by electronics and the processes leading to the vote remain unseen to the human eye. Despite the change in method of voting, the basic facets of good voting tactics remain the same: ensuring one vote per voter, maintaining voter anonymity, accuracy of vote, security of the system, and prevention of fraud.

This is where the problem lies in many arguments against electronic voting – opponents do not feel that the voting basics can be maintained in an electronic voting system. The arguments have been divided into 3 general categories of complaints: issues with the technology, vast possibilities of fraud, and protection of voters and their votes.

We'll detail the vulnerabilities in today's elections, and the viability of blockchain technology to secure the future of voting [13]. Identifying vulnerabilities in election hardware and processes is vital to preventing future attacks.

For election cybersecurity, the focus is often on hacking voting machines. However, vulnerable machines are only one part of a complex, interconnected system with multiple weak points for bad actors to exploit. Securing elections requires securing the entire process.

The five most vulnerable parts of the electoral process, outlined by Ben Buchanan and Michael Sulmeyer of Harvard's Belfer Center Cybersecurity Project, are:

- Information warfare
- Electronic voter registration databases
- Voting machinery and tabulation systems
- Election reporting systems
- Post-election audits

In addition, here comes the Blockchain. Blockchain's fundamental characteristics — transparency, immutability, and accountability — underscore the technology's potential for securing elections. In its most basic form, blockchain is a digital ledger. The technology draws its power from the peers – or nodes – on its network to verify, process, and record all transactions across the system. This ledger is never stored, but rather exists on the “chain” supported by millions of nodes simultaneously. Thanks to encryption and decentralization, blockchain's database of transactions is incorruptible, and each record is easily verifiable. The network cannot be taken down or influenced by a single party because it does not exist in one place [12].

In ideal conditions as occurs with paper ballot voting, information about whom the voter voted is kept secret. This information is not even known to the Election commission. This security aspect is very central to the electronic voting system. The system should be secure enough that no one should be able to know whom the voter voted for and tamper with it at later stages. Due to assure the highest possible security, hyper-ledger fabric is a better choice, which is actually a framework, helps to configure and use private blockchain. An ideal integration of this Framework as a Service (FaaS) is, voters' vote will validate by Trusted Third Party (TTP) and send it to election commission where administration nodes, endorsing committing peer nodes and ordering peer nodes acts as per government rules or constitution. From this level, all of the different clusters of nodes will communicate with the proposed Hyper-ledger

Fabric based FaaS, then it'll verify, validate, process and respond the output or result to end-user nodes of election commission. More physically, in this cycle, data should be travel into Micro-service layer via RESTful APIs and the micro-service layer will communicate with the hyper-ledger fabric framework to perform their specified job. All the data and services should be manage on election commissions custom infrastructure or data center which is actually a distributed database cluster using several machines.

Voting is a right for every citizen and it must be transparent. Many scenarios across the world proves the dissatisfaction of voters of voting process and some times the voting result. No matter the arena, voters demand trust. They deserve it. And here we comes with the best possible solution, proposed the most secured way of voting with the help of Blockchain using Hyper-ledger Fabric based Framework as a Service (FaaS) which can used to implement a highly maintainable, large-scale, cost effective E-voting solution.

II. RELATED WORK

Sudershan et al. [2] argue that it is possible to send thousands of transactions per second onto the blockchain using a Hyper-ledger permissioned blockchain, utilizing every aspect of the chain code to ease the load on the blockchain. The fabric allows the user defining assets from client side and using them with the Fabric Composer without any cryptocurrencies involved in the transaction.

Denis et al. [3] argue that the approach of their design allows carrying out combined voting of both traditional paper voting and e-voting. It discusses its implementation based on Hyperledger Fabric platform, describes the architecture of the solution demonstrating its functionality. Since the issue of using e-voting in both corporate and government voting has not yet been fully resolved, there remains a wide scope for improving existing approaches and proposing new protocols enabling the voting system to be resistant to various kinds of attacks. Due to the rapid development of distributed ledger technologies and their potential for solving existing problems the authors propose a modified protocol of the published earlier voting scheme which is complemented by blockchain technology to increase trust between participants. This approach allows carrying out combined voting of both traditional paper voting and e-voting.

Blockchain as a nascent technology has ushered in a range of opportunities in different sectors such as finance, real estate, supply chain management, medical records, elections etc. Ishaku et al. [4] argue that key users of the system or actors were isolated and their interactions with the system identified using Use Case Methodology. The system actors are the voters, miners that consists of Universities and public libraries, central authority, candidates and the voters. The different stages of an election process such as pre-election, registration of both candidates and voters, voting/balloting, tallying and auditing were equally enumerated. The interactions of the actors with

the system were equally captured, analyzed and used to develop the conceptual system.

Alexander et al. [5] presented some core issues in eVoting transparency as required by the ruling of the German Constitutional Court. In particular, it suggested a way to define and operationalise the terms "audit chain", "count" and "recount", which arguably represent the core issues in eVoting transparency and auditability. Furthermore, the paper introduced a model to map the key security dimensions in an eVoting system and the degree to which they are fulfilled by technical, not just organisational, means, with auditability being the dimension discussed in this contribution.

Wang et al. [6] propose a large-scale e-voting design using a permission-based protocol called Delegated Proof of Stake (DPOS). DPOS protocol can be a small network of nodes that have obtained permission from an administration authority. The paper addressed the issue of anonymity and voter privacy by using a one-time ring signature protocol. It makes use of smart contracts for recording, managing, calculating and checking votes.

III. PROPOSED FRAMEWORK AS A SERVICE

Framework as a Service is actually an intermediate stage or condition between Platform as a Service (PaaS) and Software as a Service (SaaS). In typically scenarios, it requires less work to develop and implement a system, because it has the speciality of both PaaS and SaaS. In ideal conditions, the proposed Hyper-ledger fabric based FaaS, contains several layer or level of application services within an infrastructure. Firstly, this FaaS is supposed to be platform independent, which means it will compatible with the election commission's existing data center which is actually a distributed database server clusters using several machines. All the blockchain level services, application level services and API endpoints will be host there. And this will reduce execution time and operational cost due to reusing existing components and databases of election commission.

The proposed Hyper-ledger fabric based FaaS contains three layers or level of services. They're-

- Hyper-ledger Fabric Framework
- Micro-services Layer
- RESTful APIs Layer

Each services has their own responsibilities. They are inter-related components, but they are not inter-depended. This is a fail-safe design to ensure system reliability. This can also be called micro-service architecture. All the layers is designed to assign specific job to each components to ensure efficiency, reusability, maintainability and hisgest possible security.

In ideal conditions, this FaaS based E-voting executing scenario is; from Fig. 1, the voters' vote will validate by Trusted Third Party (TTP) and send it to election commission where administration nodes, endorsing committing peer nodes and ordering peer nodes acts as per government rules or constitution. In this cycle, data should be travel into

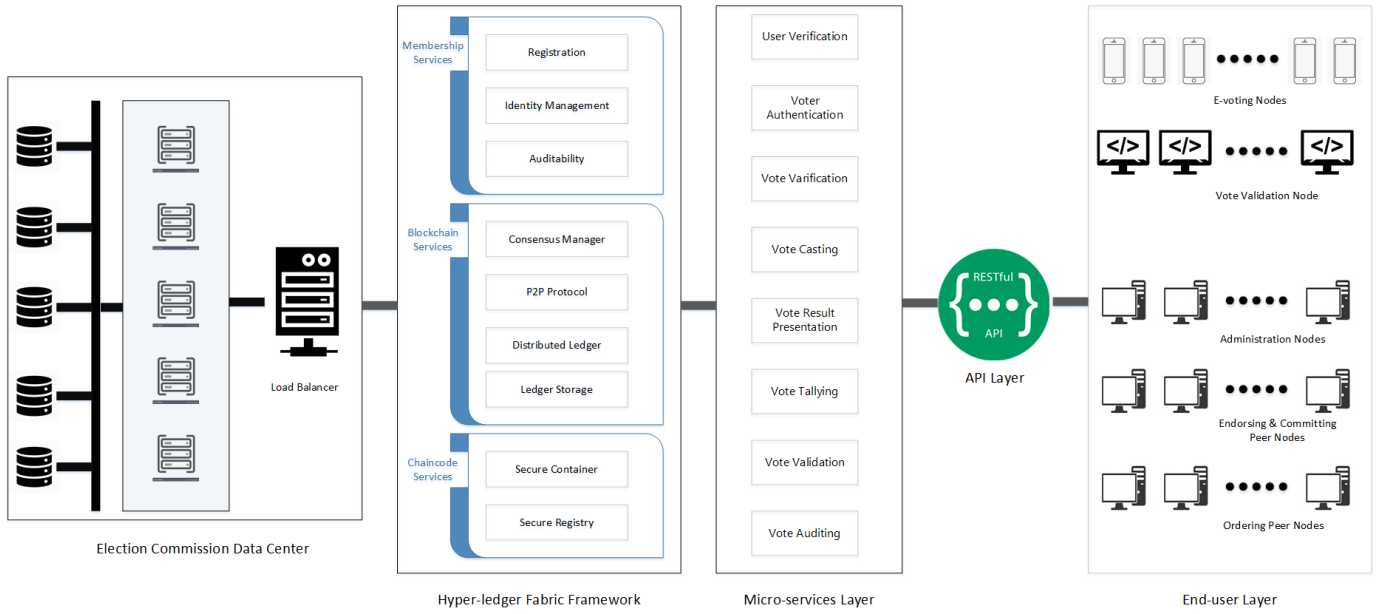


Fig. 1. A top level architectural design of a E-voting system using Hyper-ledger Fabric based Framework as a Service (FaaS).

Micro-service layer via RESTful APIs in Json data format. Micro-service layer will deserialize Json data format as regular object type and validate it first. Then it will process by using the components or class libraries of hyper-ledger fabric framework to perform their specified job and send the response into the micro-service layer. Then, from the micro-service layer further processing should be executed and send it to end-user layer via RESTful APIs. Finally after voting process ends as per election commission policy based on laws and orders, the result of election will be available to every node.

A. Infrastructure

This is actually the data center of election commission. There will be many clusters of servers. All the servers contain databases. In simple words, it can be called as distributed database server infrastructure. All the blockchain services, micro-services and RESTful APIs will be hosted into dedicated servers for each. More physically, blockchain server cluster will communicate with micro-service server clusters. The micro-service server cluster will communicate with RESTful APIs dedicated server. There will be a dedicated load balancer server to communicate with RESTful APIs server. The end-user level nodes will make request to load balancer and then the load balancer distributes the request to the server in which RESTful APIs are hosted.

B. Hyper-ledger Fabric Framework

This is the foundation which makes the e-voting solution possible, with the use of Hyper-ledger Fabric. Hyper-ledger

Fabric is a blockchain framework implementation which enables the development of blockchain information system solutions by using a modular architecture approach. The Hyper-ledger Fabric enables the creation of permissioned blockchain networks, which inherit the security, and privacy properties of underlying blockchain. In addition to peer nodes, Hyperledger Fabric network uses "Ordering nodes" which are used to ensure the consistency of the blockchain by delivering ordered blocks of an endorsed transaction to the committing peer nodes to add to the blockchain [7].

C. Micro-service Layer

Micro-services are a software development technique—a variant of the service-oriented architecture (SOA) structural style—that arranges an application as a collection of loosely coupled services. Using micro-services architecture for design and development systems, all the services become fine-grained and the protocols be lightweight [8]. This layer is used as application services which consist of all services within the e-voting system. These services perform by heading into a general objective together. These services are inter-related, but they are not inter-dependent. They enable all the access to services is governed through access control and permissions determined by the responsibilities of each node type.

D. RESTful APIs

Representational state transfer (REST) is a software architectural style that defines a set of constraints to be used for creating Web services. Web services that conform to the REST architectural style, called RESTful Web services, provide interoperability between computer systems on the Internet. RESTful Web services allow the requesting systems to access and manipulate textual representations of Web resources by

using a uniform and predefined set of stateless operations. Other kinds of Web services, such as SOAP Web services, expose their own arbitrary sets of operations [9]. Here, the RESTful APIs acts as a manager between the end-user layer and micro-service layer. It will actually receive data from end-users and send it to micro-service on json data format. This API layer helps to assure the system as more secured. In the end-user layer, voters can give their votes by using cross-platform mobile apps. The TTP will validate the voting blockchain by using specified web app of election commission. The election commission will maintain the e-voting system by their decentralized apps (dApps).

E. End User Level Decentralized Apps (dApps)

A decentralized application (DApp, dApp, Dapp, or dapp) is a computer application that runs on a distributed computing system. DApps have been popularized by distributed ledger technologies (DLT) such as the Ethereum Blockchain, where DApps are often referred to as smart contracts [10]. DApps may run on top of distributed computing systems such as Ethereum or Bitcoin. Decentralized applications are stored on and executed by a blockchain system [11]. Here, voters will give their precious vote and see result after end of election by using cross-platform mobile dApps. The Trusted Third Party use web based dApps for validating voters and votes. The election commission administration department, audit department and IT department will use desktop based dApps for completing the election process.

IV. DISCUSSION

The proposed Framework as a Service ensures an improved quality of e-voting. Firstly, the system is a highly and moderately secured blockchain based system because of its multiple layers of communication using. It is actually a FaaS, which means any government can use it by performing some small application level changes, which assures the flexibility and reusability of e-voting system. In simple words, it makes e-voting a highly maintainable system. The proposed framework is efficient because of micro-service base architecture. Using this FaaS, applications will not depend on any specific browser, machine or operating system, which ensures the larger scalability of the system. It contains integrity with any infrastructure. The ideal architecture enforces vendors to build highly reliable e-voting system. Finally, it can be state that, using this proposed Framework as a Service, a blockchain based e-voting system will be a highly efficient, maintainable, reliable and secured large-scale solution.

V. CONCLUSION

Voting is the core right of every citizen. It must be transparent. Many scenarios across the world proves the dissatisfaction of voters of voting process and sometimes the voting result. No matter the arena, voters demand trust. They deserve it. From this scenario, the objective becomes clear to design a solution by which a fair and transparent election can be executed with the minimal possible execution and operational cost. Proposed

Hyper-ledger Fabric based Framework as a Service (FaaS) can be used for any government, and any level of election. It has integrity with existing election commission's data center. Government has flexibility to customize E-voting system using this FaaS, which means budget, language, racial or religious barrier can not hamper the electoral process. Using this FaaS, vendors are directed to develop end-user level dApps by following software engineering best practices. This FaaS not only maintainable to itself, but also enforces vendors to assure highly maintainable deliverable for end-user level dApps. The proposed Hyper-ledger Fabric based Framework as a Service (FaaS) also assures the top most quality by covering almost all the possible quality factors. The proposed model will be successful when voters will get a fair and transparent election by using this FaaS based E-voting system. Because technology is born for human, and we got blockchain.

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